

Abnormal Molecular Mass

- (c) Na_3PO_4 gives maximum four ion it is show highest Vant's haff factor.
- (a) $K_4[Fe(CN)_6]$ dissociates as $4K^+ + [Fe(CN)_6]^{4-}$, thus 1 molecule dissociates into five particles in the similar way $Al_2(SO_4)_3$ also gives five particles per molecule.
- (a) Benzoic acid in benzene undergoes association through intermolecular hydrogen bonding.

4. (c) von't's Hoff factor (i) = $\frac{\text{experimental C.P.}}{\text{Calculated C.P.}}$
 $= 1 - \alpha + x\alpha + y\alpha$, for KCl it is = 2 and for sugar it is equal to 1.

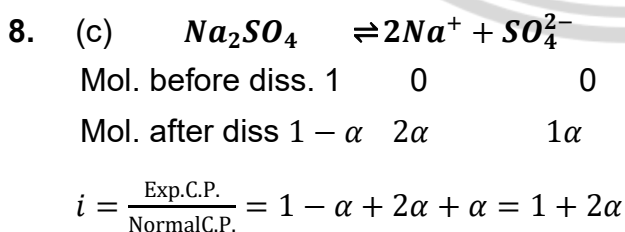
5. (c) $Ca(NO_3)_2 \rightarrow Ca^{2+} + 2NO_3^-$ it gives three ions hence the Van't Hoff factor = 3.

6. (a) $m = \frac{5 \times 18 \times 2.5}{0.04 \times 80} = 70.31$

7. (a) **Less**

Explanation:

- In association, molecules combine together, decreasing the number of particles, so van't Hoff factor $i < 1$.
 - In dissociation, particles split into ions, increasing the number of particles, so $i > 1$.
- Therefore, the van't Hoff factor for association is always less than that for dissociation.



9. (a)

Explanation:

Van't Hoff factor:



$$i = \frac{\text{Normal molar mass}}{\text{Observed molar mass}}$$

Also:

- $i > 1$ for dissociation
- $i < 1$ for association

Therefore:

- (c) is wrong
- (d) is wrong

Only (a) is correct.

10. (d) $MgSO_4$ dissociates to give 2 ions.

11. (d) Urea does not give ion in the solution.

12 **Explanation:**

Suppose degree of association = x

Initially:

1 mole

After association:

- Unassociated moles:

$1 - x$

Associated moles formed:

$\frac{x}{n}$

Total moles after association:

$$1 - x + \frac{x}{n}$$

Van't Hoff factor:

$$i = 1 - x + \frac{x}{n}$$

(c)



13. (b) Molecular weight of $CH_3COOH = 60$

Hence the molecular weight of acetic acid in benzene = $2 \times 60 = 120$.

- 14 (b) Benzoic acid molecules are associated in benzene

Explanation:

In benzene, benzoic acid forms dimers by hydrogen bonding. Association decreases the number of particles, so osmotic pressure becomes lower than expected.

- 15 (b) Greater than 1

Explanation:

Electrolytes dissociate into ions, increasing the number of particles.

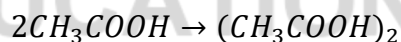
$$i > 1$$

Hence observed molar mass becomes smaller than normal molar mass.

- 16 (a) Water prevents association of acetic acid

Explanation:

- In water, acetic acid exists mainly as single molecules $\rightarrow$ molar mass = 60
- In benzene, acetic acid associates to form dimers:



Hence observed molar mass becomes nearly double (120).

17. (b) $AlCl_3$ furnishes more ions than $CaCl_2$ and thus shows higher boiling point i.e. $t_1 > t_2$.

18. (d) $Na_3PO_4 = 3Na^+ + PO_4^{3-}$.

19. (b) Benzoic acid dimerises due to strong hydrogen bonding.

